Cooperative Tradeoff Analysis of Consortia for Plant-Based Fermentation

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Abstract. The global growth in plant-based food consumption has made chickpea and fava bean strong candidates for replacing animal protein due to their nutritional profile. However, the presence of off-flavors - including herbaceous, bitter and beany notes - has limited their acceptance among consumers. These sensory attributes arise from complex biochemical pathways, including lipid oxidation mediated by lipoxygenase and protein degradation. Traditional empirical screening is insufficient to address this challenge, as it overlooks the metabolic dynamics of multi-strain communities. This work proposes the use of Genome-Scale Metabolic Models of selected lactic acid bacteria, analyzed by Flux Balance Analysis, to computationally model the behavior of microbial consortia in legume matrices. Using community modeling tools such as COBRApy, SMETANA and MICOM, different simulation approaches were evaluated with the aim of predicting the mitigation of off-flavors and identifying optimized simulation conditions.

Keywords: Biological Modeling · Lactic Acid Bacteria · Microbial Consortia · Genome-Scale Metabolic Models · Fermentation.

1 Introduction and Objectives

Legumes, particularly chickpea (*Cicer arietinum*) and fava bean (*Vicia faba*), are nutritionally dense sources of vegetable protein, fiber, vitamins and minerals, making them strong candidates for replacing animal protein in plant-based food systems [1,2,3]. However, consumer acceptance is limited by a persistent sensory obstacle: the presence of undesirable aromas collectively referred to as off-flavors, including beany and grassy notes, bitterness and earthy aromas [4]. These attributes result from three main biochemical pathways: lipoxygenase-mediated lipid oxidation [5,6]; Ehrlich pathway and Strecker degradation [7,8]; and the accumulation of other volatile compounds from the Maillard reaction [9].

The conventional approach to mitigating these compounds relies on empirical screening, which presents significant and well-documented limitations: high costs, biological and experimental variability, and a lack of mechanistic understanding [9,10]. These limitations are particularly relevant in multi-strain microbial communities, where interactions such as cross-feeding and substrate competition shape fermentation outcomes in ways that cannot be predicted from single-strain behavior [11]. Although microbial consortia offer functional advantages, including greater stability and metabolic complementarity, their rational design remains a challenge [11,12,13]. Genome-scale metabolic modeling emerges as a promising tool. The simulation of microbial consortia can follow different strategies, from steady-state approaches to dynamic models, depending on the biological question, the quality of available models and the available experimental data [11]. The present project aims to computationally model lactic acid bacteria (LAB) consortia in chickpea and fava bean matrices in order to predict off-flavor mitigation and identify optimized fermentation conditions. The following objectives were defined: (i) curate and standardize GSMMs of 2-3 candidate LAB strains in SBML format; (ii) define medium conditions representative of legume matrices in COBRApy; (iii) analyze potential metabolic interactions and cross-feeding in the consortium using SMETANA; (iv) implement community simulations with MICOM to predict metabolic fluxes and community growth under different strain abundance configurations; (v) evaluate off-flavor mitigation potential and community sensitivity to medium perturbations.

2 State of Art

The undesirable sensory profile of chickpea and fava bean results from the presence of volatile compounds generated by biochemical pathways native to the plant [4,5]. The most studied pathway is lipid oxidation mediated by lipoxygenase (LOX), which catalyses the conversion of polyunsaturated fatty acids into hydroperoxides, subsequently cleaved into short to medium chain volatile aldehydes such as hexanal and (E)-2-nonenal, responsible for grassy and rancid notes [5,6]. Simple hydration or grinding of the legumes is sufficient to trigger this cascade, meaning these off-flavors originate intrinsically within the plant [4]. Protein degradation and amino acid catabolism through the Ehrlich pathway and Strecker degradation generate compounds such as 3-methylbutanal and phenylacetaldehyde [5,7], while the Maillard reaction contributes sulphurous and roasted volatiles [9]. Heat treatments, even though they are efficient in inactivating LOX, compromise protein functionality and nutritional value [3], justifying the use of fermentation as an alternative strategy.

Lactic fermentation is a widely explored strategy for improving the sensory profile of legumes [10,14], that preserves protein functionality and nutritional profile of the matrix, constituting a technologically relevant and effective alternative [2,3]. LABs contribute to off-flavor mitigation through two main mechanisms: indirectly, by producing lactic acid, which lowers pH and inhibits LOX activity - optimal between pH 6.0 and 6.4 [6,10]; and directly, through enzymatic action, in which Ehrlich pathway intermediates are converted intracellularly to acids and alcohols before secretion, rather than accumulating as volatile aldehydes [14]. Among the LAB species with the most documented experimental evidence in legume fermentation, *Lactiplantibacillus plantarum* stands out for its efficient acidification capacity and extensive proteolytic system, with functional ADH enzymes directly associated with hexanal reduction [6,10,15]. *Leuconostoc mesenteroides* is an equally relevant candidate, with demonstrated capacity to reduce volatile aldehydes in legume matrices via its heterofermentative metabolism [4,6,16,17]. *Pediococcus pentosaceus* has also been reported in legume fermentation contexts [4], though the suitability of available genome-scale models for community simulation requires evaluation. Additionally, some LABs produce positive aroma compounds that can mask residual off-flavors and enrich the final sensory profile [18]. To move beyond empirical screening, Genome-Scale Metabolic Models (GSMMs) are computational tools that integrate genomic, biochemical and physiological data, establishing the link between genotype and phenotype [19,20]. These models represent the totality of an organism's metabolic reactions and allow predictions of growth rates and metabolic behavior under defined environmental conditions [21]. These predictions are typically obtained through FBA, which formulates a linear programming problem: maximizes an objective function - usually biomass growth - subject to steady-state and the flux limits of each reaction [20]. Model quality is critical, as incomplete annotation produces unreliable predictions [11]. Curated GSMMs exist for the primary candidate species: iLP728 for *L. plantarum* WCFS1 and the Koduru2022 model for *L. mesenteroides* ATCC 8293, both publicly available [16,22]. GSMMs for these organisms have recently been applied to the study of microbial consortia, enabling the prediction of cross-feeding and substrate competition [11,23]. The modeling of microbial consortia using GSMMs can follow different strategies depending on the biological question and model quality [11]. Scott et al. [11] evaluated constraint-based community modeling tools and found MICOM, SteadyCom and cFBA among the top-performing tools for two-species consortia across qualitative and quantitative criteria; individual model quality was identified as the primary limiting factor across all approaches. SteadyCom and cFBA were not implemented here, as MICOM and SMETANA already cover the predictive needs of a two-species consortium. MICOM implements a two-level optimization approach - maximizing community growth first and then individual member growth - and is particularly suitable for predicting cross-feeding and stable relative abundances [11,24]. SMETANA performs static analysis of potential metabolic exchanges between pairs of species, calculating interaction scores that identify likely cross-feeding candidates prior to full simulation [23]. However, it is worth noting that SMETANA's MILP-based framework can encounter structural convergence limitations when applied to organisms with extensive auxotrophies, such as LAB, where the minimal uptake set computation may not converge under calibrated rich-medium conditions - as noted in the tool's implementation [11,23].

3 Problem Description

No published pipeline combines LAB GSMMs with a computational medium nutritionally faithful to a real legume matrix to simultaneously predict metabolic interactions between species and off-flavor mitigation capacity. Three specific gaps account for this absence. First, the publicly available GSMMs for LABs have not been built or validated for legume fermentation media: iLP728 was developed under standard laboratory conditions; Koduru2022 presented zero growth in its original form due to internal identifier inconsistencies. Second, defining a computational medium that accurately reflects the nutritional composition and bioavailability of chickpea and fava bean is unprecedented in LAB modeling literature, requiring a new pipeline to convert nutritional data into exchange reaction bounds. Third, the harmonization of namespaces between models of different organisms is a prerequisite for community model construction: identifier mismatches between iLP728 and Koduru2022 would generate separate substrate pools per organism instead of a shared medium, invalidating any competition or cross-feeding analysis. The current project addresses these three gaps directly.

4 Methodology

All phases were implemented in Python 3.10, using COBRApy 0.30.0, reframed 1.6.0, SMETANA 1.2.1 and MICOM 0.39.0 as main dependencies [23,25,26,27,28]. The solver used was Gurobi with an academic license [29].

4.1 GSMMs' Curation

The models obtained for each LAB - iLP728 for *L.plantarum* WCFS1 [15] and Koduru2022 for *L.mesenteroides* ATCC 8293 [16] - were imported with COBRApy and submitted to quality analysis, which included memote evaluation [30], gene / metabolite / reaction counts, growth rate verification in default medium, biomass escape tests and FVA for blocked reaction quantification.

For iLP728, auxotrophy diagnosis was performed by selectively enabling exchange of vitamins, nucleobases and amino acids to identify which nutrient classes constrained growth. Off-flavor pathway additions included hexanal and (E)-2-nonenal as intra and extracellular metabolites with passive diffusion transport, alcohol dehydrogenase (ADH) reactions reducing each aldehyde to its corresponding alcohol (putatively assigned to lp_3054), and exchange reactions for both aldehydes and alcohols. Strecker aldehyde exports were verified; isobutyraldehyde was not added, as no KDC reaction for valine catabolism is reconstructed in the model.

For Koduru2022, zero growth in the original model was investigated via biomass component sensitivity analysis. Namespace harmonization converted (e) suffixes to _e following BiGG conventions; a glucose identifier correction (glc-D_e to glc__D_e across five reactions) ensured a shared glucose pool in the community model. Off-flavor exchanges were added following the same rationale as iLP728.

Two additional LAB candidates were evaluated for legume medium compatibility: the AGORA2 model of *Pediococcus pentosaceus* ATCC 25745 [31] and the *Streptococcus thermophilus* LMG 18311 model [32]. Both were excluded based on growth and pathway criteria detailed in Section 5.1.

4.2 Legume Medium

In order to define the legume medium, the nutritional composition of *Cicer arietinum* and *Vicia faba* was extracted from two food databases - FRIDA, USDA FoodData - and complemented with published reviews [33,34,35,36]. Following this literature review, discrepancies were observed in several nutrient categories across the sources. To address this, an interval approach was implemented: each nutrient was assigned a minimum, median and maximum value derived from the reported range, resulting in three compositional scenarios for each matrix. The maximum scenario was used as the canonical reference for all subsequent SMETANA and MICOM simulations, as it reflects the upper bound of nutritional availability.

To convert the nutritional values into FBA exchange bounds ($mmol/gDW/h^{-1}$), the Sauer batch-culture framework was used as a reference [37]:

$$q_S = \frac{\Delta S \cdot \mu}{\bar{X}}$$

where ΔS represents the change in substrate concentration ($mmol/L$), $\bar{X}$ the biomass concentration and μ the specific growth rate (h^{-1}). As no experimental batch fermentation data for these LAB strains in legume matrices fermentation is available, an approximation was made - assuming near-complete substrate consumption ($\Delta S = S_0$) and substituting μ = 1/t:

$$qS = \frac{S_0}{\bar{X} \cdot t}$$

with $\bar{X} = 0.5\ gDW/L$ (representative inoculation density for batch LAB fermentation, consistent with [22,38]), S_0 being the initial nutrient concentration in the medium ($mmol/L$) and $t = 24\ h$ (duration of the fermentation). A hydration ratio of 5 L medium per kg dry flour was used to convert mg/kg DW to mg/L [39].

The different bioavailability of each nutrient category was incorporated through the definition of a specific efficiency fraction: 1.0 for free sugars, vitamins and minerals; 0.90 for disaccharides; 0.20 for amino acids derived from protein hydrolysate tables; 0.12 for lipids; and strain-specific values for raffinose-family oligosaccharides (0.55 for *L.plantarum*, 0.60 for *L.mesenteroides*), reflecting documented differences in saccharolytic capacity [40,41,42,43,44].

The exchange bounds were applied in three successive layers. First, universal inorganic compounds that are present in any aqueous fermentation medium, but absent from the macronutrient composition, were opened unlimitedly. Second, auxotrophic components for LABs - nucleobases (adenine, guanine, uracil, thymidine, hypoxanthine, inosine) and pyridoxamine - were enabled at -1 $mmol/gDW/h^{-1}$. In the final layer, bounds derived from the nutritional composition were implemented and for any compound present in both layers, these composition-based limits took precedence over those from the supplement layer.

4.3 SMETANA Analysis

SMETANA community model was built using the reframed Community Class, with merge_extracellular_compartments=False to preserve each organism's internal namespace while sharing a common external compartment via pool reactions. Prior to the construction of the community model, the Koduru2022 model underwent in-memory ID harmonization: reframed encodes hyphens in metabolite IDs as __45__ (ASCII code for the hyphen character), producing IDs such as gln__45__L_e instead of the BiGG-standard gln__L_e. Renaming 115 such IDs back to the double-underscore convention was necessary because the legume medium lookup uses BiGG IDs exclusively. Without this step, duplicate pool reactions would block *L.mesenteroides* from approximately 115 nutrients. Therefore, the resulting merged model comprised 2073 reactions and 1773 metabolites with 173 shared pool reactions mediating metabolite exchange between the two namespaces.

Afterwards, the community underwent validation through eight structural checks: organism count, the presence of R_Community_Growth and Sink_biomass reactions for both strains, coverage of pooled reactions for all compounds in the legume medium maps and the absence of residual pools. In addition, basal growth in unrestricted whole medium was confirmed for both organisms. Community-level uptake limits were established using MAX_EFFICIENCY, which represents the maximum efficiency per compound between both LABs, ensuring that pooled reactions accurately reflect the availability of substrates from both strains. Individual exchange limits specific to each strain were applied separately.

SMETANA implements several algorithms to analyze cross-feeding interactions: Metabolite Resource Overlap (MRO) and Metabolite Interaction Prediction (MIP) characterize community-level nutritional overlap and potential exchange; Species Coupling Score (SCS), Metabolite Uptake Score (MUS) and Metabolite Production Score (MPS) characterize individual pairwise dependencies [23]. Scores were calculated across six scenarios (chickpea and fava bean $\times$ min/med/max), with the minimum growth rate defined as 10% of the slower organism's individual rate.

4.4 MICOM Community Simulations

Before building the community, both curated models were pre-processed to ensure compatibility with MICOM's objective mapping. The community was constructed from a taxonomy table specifying organism IDs, initial abundances and models. The legume medium was applied directly using the same approximation of the Sauer equation and efficiency fractions.

Two matrices were prepared - chickpea and fava bean at the maximum compositional scenario. Solo COBRApy growth rates on the same medium served as reference baselines for all subsequent comparisons.

In order to simulate the community, the cooperative_tradeoff objective was used with fraction = 0.5, meaning that each organism achieved at least 50% of its theoretical maximum growth rate simultaneously [25]. Three relative abundance scenarios were tested per matrix: *L.plantarum* dominant (75:25), equal representation (50:50) and *L.mesenteroides* dominant (25:75). Tradeoff sensitivity was assessed at fraction values of 0.3, 0.5, and 0.7 - spanning a range from a more competitive regime (0.3, each organism guaranteed only 30% of its individual maximum) to a more cooperative one (0.7, 70% guaranteed), bracketing the equal-weight default (0.5) used in the main analysis.

Off-flavor metabolite trackability was assessed by mapping Ehrlich pathway products, hexanol, phenol, acetoin and ethanol to community exchange reactions. Although hexanal uptake and ADH reduction were represented in both GSMMs, neither model contains a LOX reaction to generate it endogenously; risk was therefore assessed indirectly via linoleic acid (its precursor) availability. To confirm *L.mesenteroides*'s Ehrlich pathway capacity independently of the community objective, parsimonious FBA (pFBA) was run on each model individually under both legume media.

Additionally, three medium perturbations were performed across all abundance configurations and both matrices - sucrose and maltose removal, 50% reduction of amino acid uptake bounds and anaerobic condition. The same perturbations were applied under solo COBRApy FBA to isolate strain-specific sensitivities masked by the cooperative tradeoff objective.

5 Results and Discussion

5.1 GSMM Curation

Both models were successfully curated and demonstrated growth on both legume mediums. iLP728's initial growth of nearly zero ($\mu = 0.0025\ h^{-1}$) was not a structural defect, but rather a result of overly restrictive exchange conditions incompatible with *L.plantarum*'s auxotrophic profile. Enabling specific vitamin, nucleobase and amino acid exchanges restored growth to $\mu = 0.2116\ h^{-1}$, aligning with the experimentally reported ranges [22,31]. The curation included adding the complete representation of the LOX pathway and confirmed presence of Strecker aldehyde exports. Isobutyraldehyde was excluded due to the absence of a reconstructed KDC reaction for valine - a previously documented limitation of the model.

For Koduru2022, biomass component sensitivity analysis identified peptido1_LME_c as the sole critical gap: a hyphen/underscore inconsistency between the identifier produced by PPTGS (peptido1-LME_c) and consumed by SDADACP (peptido1_LME_c) broke the peptidoglycan biosynthesis chain. Renaming the producing metabolite resolved the inconsistency and restored growth to $\mu = 0.5589\ h^{-1}$, confirming that the metabolic network remained structurally intact. The final curated models are summarized in Table 1.

Koduru2022 was selected over iLME620 because the complete absence of GPR associations in iLME620 would have made gene-level analysis meaningless and its community modeling behavior uncertain. The blocked reaction rate of 40.6% in Koduru2022 v2 exceeds 26.9% in iLP728 v2, reflecting the difference in network size rather than curation quality.

The model of *P.pentosaceus* ATCC 25745 showed zero growth on legume medium due to dependency on dipeptide nitrogen sources absent from the free amino acid-based matrix. *S.thermophilus* LMG 18311 grew on chickpea ($\mu = 0.0347\ h^{-1}$) but not on fava bean ($\mu = 0.000\ h^{-1}$), using galactose as its sole effective carbon source; all Ehrlich product exchange reactions were absent from the model. Both candidates were excluded.

Table 1. GSMMs curation summary. Blocked reactions (%) and growth rate (h^{-1}) reported for each model version.

Model	Version	Reactions	Metabolites	Genes	Blocked (%)	Growth (h^{-1})
iLP728	original	771	662	728	25.6	0.0025
iLP728	curated	785	672	729	26.9	0.0025
Koduru2022	original	1094	915	687	59.8	0.00
Koduru2022	curated	1112	926	687	40.6	0.5589

5.2 Legume Medium and Solo Growth

Both strains exhibit growth on chickpea and fava bean across all three compositional scenarios - minimum, medium and maximum - which confirms the defined bounds produce biologically viable mediums for both models (Table 2).

Table 2. Solo growth rates (μ, h^{-1}) for *L.plantarum* and *L.mesenteroides* under three compositional scenarios for each legume matrix. Min., Med. and Max. refer to the minimum, median and maximum nutrient values from the literature-derived ranges.

Organism	Matrix	Min.	Med.	Max.
L.plantarum	chickpea	0.3088	0.4751	0.5645
L.plantarum	fava bean	0.0682	0.2047	0.3411
L.mesenteroides	chickpea	0.0478	0.0777	0.1076
L.mesenteroides	fava bean	0.0509	0.0716	0.0907

In the maximum scenario, *L.plantarum* achieves a growth rate of $\mu = 0.5645\ h^{-1}$ for chickpea and $\mu = 0.3411\ h^{-1}$ for fava bean, while *L.mesenteroides* reaches $\mu = 0.1076\ h^{-1}$ and $\mu = 0.0907\ h^{-1}$, respectively. The observed growth advantage of *L.plantarum* over *L.mesenteroides* in both matrices reflects its broader metabolic capacity, which aligns with *L.plantarum*'s documented faster acidification kinetics in legume fermentation [10,14]. The lower growth rates of *L.mesenteroides* are consistent throughout all scenarios and are reproduced in both SMETANA and MICOM frameworks, pointing to genuine metabolic constraints rather than calibration issues. These values are used as the reference baselines for all subsequent consortium comparisons.

5.3 SMETANA Interaction Analysis

MIP scores varied between 5 and 10 across different scenarios, suggesting a moderate potential for metabolite exchange to decrease dependence on external resources. MRO ranged from 0.29 to 0.47 in chickpea and from 0.38 to 0.53 in fava bean, meaning that competition increases with nutritional richness in chickpea, but peaks at the median compositional scenario in fava bean.

An SC value of 1.0 for *L.plantarum* does not indicate an obligatory biological dependence on *L.mesenteroides*, it reflects the results of SMETANA's MILP sampling conducted in an environment where the concentrations of pantothenate and niacin are below the feasibility threshold for *L.plantarum.* In every sampled solution, *L.plantarum* requires an external supplier, which is always *L.mesenteroides*, in the context of the merged model topology. As demonstrated by the solo FBA results, *L.plantarum* grows independently when these compounds are supplied at maintenance levels. SC = 0.0 for *L.mesenteroides* is a confirmation that this bacteria does not require *L.plantarum* metabolic outputs to grow across the entire perturbation space.

Table 3. SMETANA scores across legume medium scenarios. MIP: number of exchangeable metabolites; MRO: metabolic resource overlap; SC: species coupling score; N_MUP: number of metabolites with MUP > 0 after filtering; MPS: metabolite production score (total secretable metabolites).

Scenario	MIP	MRO	SC Lp→Lm	SC Lm→Lp	N_MUP	MPS
Chickpea (min)	6	0.2857	1.0	0.0	20	64
Chickpea (med)	10	0.3529	1.0	0.0	35	65
Chickpea (max)	10	0.4706	1.0	0.0	30	66
Fava Bean (min)	5	0.3750	1.0	0.0	21	66
Fava Bean (med)	5	0.5333	1.0	0.0	32	66
Fava Bean (max)	5	0.4000	1.0	0.0	33	66

The MUP heatmap presented in Figure 1 identifies the compounds that contribute to *L.plantarum's* dependency. Specifically, isoleucine (Ile), phenylalanine (Phe), valine (Val), niacin and pantothenate exhibit a MUP score of 1.0 in almost every scenario. On the other hand, methionine (Met) and cysteine (Cys) demonstrate highly variable MUP scores ranging from 0.48 to 1.00 and from 0.27 to 0.99, respectively. Nonenol is also identified, with a non-zero MUP in several scenarios, being the only compound relevant to off-flavors detected in the MUP analysis. This finding suggests that *L.plantarum* ability to assimilate

LOX-pathway alcohols secreted by *L.mesenteroides* is a predicted interaction rather than an assumption derived from modeling.

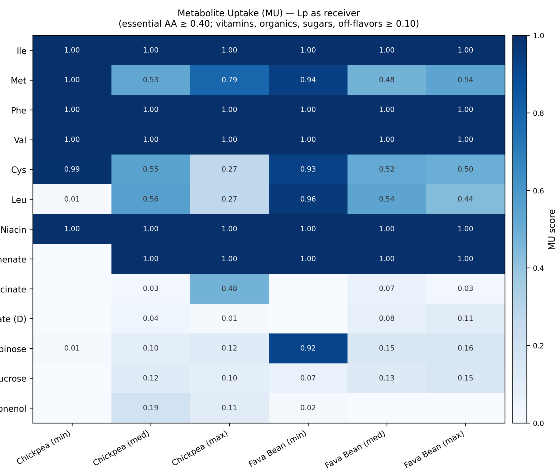

Fig. 1. MUP scores for *L.plantarum* as receiver across six legume medium scenarios. Values represent the fraction of feasible solutions requiring uptake of each metabolite. Empty cells indicate MUP below threshold or metabolite absent from that scenario.

5.4 MICOM Community Simulations

All six abundance configurations were solved optimally. A sensitivity analysis of the tradeoff at fractions of 0.3, 0.5 and 0.7 demonstrated that absolute community growth rates scale linearly with the chosen fraction, without altering the qualitative *L.plantarum*/*L.mesenteroides* asymmetry, confirming the results are not artefacts of a degenerate optimum, though the fraction remains a tractability parameter rather than a biological one.

The growth rates obtained under cooperative_tradeoff are proportional with abundance ratios by mathematical design - consequently, these findings should not be interpreted as biological results. As illustrated in Figure 2, the relevant data resides in the deviation from the solo baseline.

Regardless of the medium, *L.mesenteroides* benefits substantially from the community, surpassing its isolated growth rate. In contrast, *L.plantarum* growth is generally inhibited by the presence of its competitor, strongly depending on a greater abundance to achieve growth rates comparable to its solo baseline.

Cross-feeding analysis detected 52 interactions between community strains through all six configurations (Figure 3). The dominant fluxes primarily occur from *L.mesenteroides* to *L.plantarum*, involving (E)-2-nonenol, 2-methylpropanoate, glutamate and 2-methylbutanoate. Conversely, (E)-2-nonenal flows in the opposite direction, from *L.plantarum* to *L.mesenteroides*. Among the cross-feeding

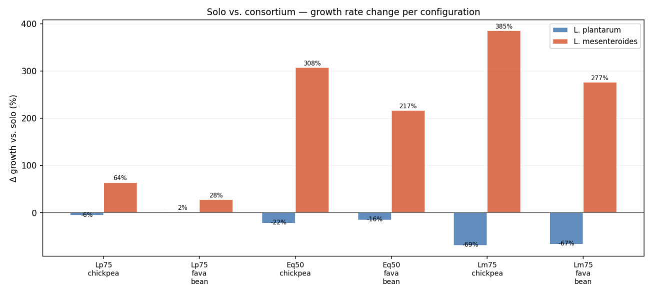

Fig. 2. Relative change in growth rate ($\Delta\%$, consortium vs. solo) for *L.plantarum* and *L.mesenteroides* across all abundance configurations and matrices. Positive values indicate growth benefit relative to solo baseline; negative values indicate penalty.

metabolites identified by MICOM, three *L.mesenteroides* to *L.plantarum* exchanges: 2-methylpropanoate, 2-methylbutanoate and 3-methylbutanoate - were independently flagged as MU-positive uptake compounds by SMETANA, providing convergent evidence across the two frameworks for these specific exchanges. The coupled exchange of LOX pathway metabolites - *L.plantarum* secreting (E)-2-nonenal that *L.mesenteroides* takes up and converts to (E)-2-nonenol, which *L.plantarum* then recovers - represents a cooperative LOX pathway interaction that could not be predicted from single-strain analysis.

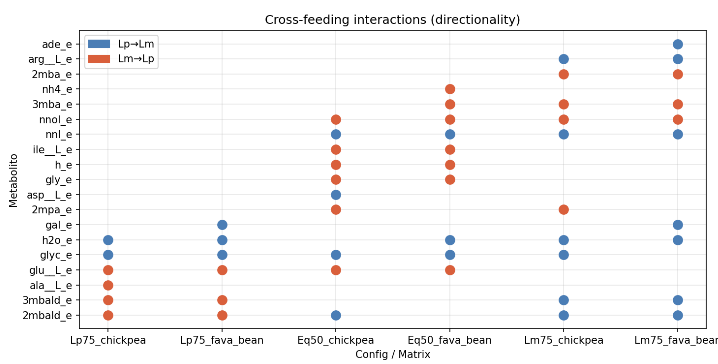

Fig. 3. Cross-feeding interactions detected in MICOM community simulations across all abundance configurations and matrices. Each dot indicates a net flux above threshold between the two organisms; blue = *L.plantarum*(Lp)→*L.mesenteroides*(Lm), orange = *L.mesenteroides*(Lm)→*L.plantarum*(Lp).

5.5 Off-Flavor Mitigation Capacity

Solo pFBA confirms *L.mesenteroides* secretes both 3-methylbutanoate (EX_3mba_e $= 0.588\, mmol/gDW/h^{-1}$) and 2-methylbutanoate (EX_2mba_e $= 2.909\, mmol/gDW/h^{-1}$) exclusively on fava bean, with no aldehyde intermediate accumulating in either route. The two products arise through distinct mechanisms: 3-methylbutanoate reflects genuine leucine transamination, consistent with classical Ehrlich-pathway overflow [8], whereas 2-methylbutanoate is predominantly (94.3%) sourced from diversion of isoleucine-biosynthetic flux rather than catabolism of dietary isoleucine - so the overflow framing holds fully only for the leucine-derived product. The valine-derived analogue (2-methylpropanoate) was not secreted in any scenario. *L.plantarum* showed no activity on this route in either matrix. Hexanal risk was assessed indirectly via linoleic acid availability, since neither GSMM can generate it endogenously.

Together, these results point to three complementary mitigation mechanisms: one intrinsic to *L.mesenteroides* metabolism, demonstrated independently of community context via solo pFBA and two that emerge specifically from inter-species interaction. *L.mesenteroides* converts BCAA intermediates to acids intracellularly before secretion, avoiding volatile aldehyde accumulation - the primary predicted mitigation route [15]. The dominant cross-feeding interaction ((E)-2-nonenol) positions *L.plantarum* as a secondary sink for LOX-pathway alcohols secreted by *L.mesenteroides*. Finally, linoleic acid consumption by both strains reduces the substrate pool available for plant LOX activity independently of microbial flux.

6 Conclusions

This work demonstrates that a GSMM-based approach can be applied to a nutritionally defined legume fermentation context to predict metabolic interactions and off-flavor mitigation capacity at consortium level - a combination absent from prior literature. The curated models for *L.plantarum* WCFS1 and *L.mesenteroides* ATCC 8293 both support growth on chickpea and fava bean media derived from real nutritional data. The Sauer-approximated medium construction with compound-specific efficiency fractions produced biologically viable bounds without requiring experimental calibration. SMETANA identified pantothenate and niacin as the primary structural dependency in the consortium. MICOM community simulations across six abundance configurations consistently show *L.mesenteroides* benefiting from the consortium while *L.plantarum* is penalized under competitive conditions, with chickpea supporting higher overall growth rates. Three off-flavor mitigation mechanisms emerge: *L.mesenteroides*'s intracellular Ehrlich processing of BCAA intermediates before secretion, cross-feeding of (E)-2-nonenol from *L.mesenteroides* to *L.plantarum* and linoleic acid depletion by both strains reducing the substrate available for plant LOX activity.

The main limitation is the steady-state nature of the cooperative tradeoff objective, which cannot capture temporal dynamics or the pH-mediated LOX inhibition that drives fermentation outcomes in practice. Cross-feeding fluxes were extracted from a single optimal MICOM solution and were not verified by flux variability analysis; their uniqueness under alternative optima remains untested. Additionally, iLP728 lacks calibrated non-growth-associated maintenance, which inflates absolute growth rates without affecting relative strain or matrix comparisons, and the cooperative_tradeoff fraction is a tractability choice rather than a biological parameter - varying it from 0.3 to 0.7 rescales absolute community growth by roughly ±40% without altering the qualitative *L.plantarum*/*L.mesenteroides* asymmetry reported here. Experimental validation of predicted cross-feeding interactions, particularly pantothenate transfer and (E)-2-nonenol uptake by *L.plantarum*, remains the necessary next step.

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